

# Introduction to Coding Theory HW 3

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(1)

a)

We have  $C$  is given by  $F(X, Y, Z) = X^{q+1} - Y^q Z - Y Z^q = 0$ .

First, consider case  $Z = 0, Y = 1$ . We have  $F_X = (q+1)X^q = 0 \implies X^q = 0$ . We have  $F_Z = -Y^q - qYZ^{q-1} = -Y^q = -(1)^q \neq 0$  so we never have a singular point in the affine plane.

Consider next case  $Z = 1$ . We then have  $F_Z = -Y^q - qYZ^{q-1} = -Y^q - qY$  and similarly  $F_Y = -qY^{q-1} - Z^q$ . At any singular point We have already  $X^q = 0$  so we have  $F(X, Y, 1) = X^q X - Y^q - Y = 0 \rightarrow -Y^q - Y = 0$ . So, we have  $Y(Y^q + 1) = 0$ . If we assume  $Y = 0$  then we get  $F_Y = 0 - Z^q = 1$ , which can not occur, so we have  $Y^q = -1$ . Though we then have  $F_Z = 1 - qY = 0$  so we have  $Y = \frac{1}{q}$  so we have  $F\left(X, \frac{1}{q}, 1\right) = 0 + 1 - \frac{1}{q} = 0 \rightarrow q = 1$ , though we have  $q$  is prime so this is a contradiction, so there are no singular points on  $C$  so  $C$  is nonsingular.

b)

First consider affine points with  $Z = 1$ . These then must satisfy  $X^3 = Y^2 - Y$ . We have  $X^3 \in \{0, 1, 2^3 = 0, 3^3 = 3\}$  and similarly  $Y^2 - Y \in \{0, 0, 2^2 - 2 = -$